Yelp Recommendation Systems & Business Churn Prediction

Tampa Bay | Restaurants, Bars & Coffee Shops

MAS 651 — Machine Learning II | Spring 2026

Luna Gerlic Santoni, Daniel Regalado, Jeanne Hassoun
University of Miami

Agenda

- Data & Tiered Strategy
- Item-to-Item Recommendations
- User-to-Business Collab. Filtering
- Churn Prediction & Survival Analysis
- Business Insights
- Assumptions & Limitations

Data & Tiered Strategy

Yelp Open Dataset

150K businesses nationwide

7M reviews

8,012 Tampa Bay restaurants/bars/coffee

99.6% base sparsity

Three Tiers, One Dataset

Tier	Purpose	Users	Biz	Sparsity
T1 Full	Churn + EDA	22,657	3,769	99.62%
T2 Med	Item-to-Item	7,352	2,218	98.83%
T3 Dense	SVD + NCF	2,279	1,390	96.82%

7x density improvement for collab. filtering vs. single-tier approach

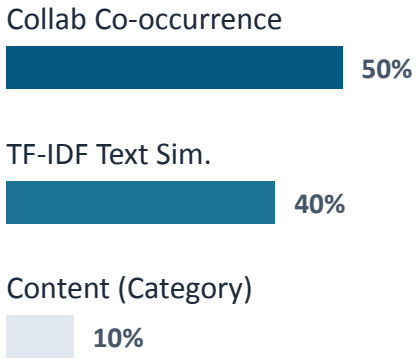
Train/Test: per-user time-based split (last review = test). Zero leakage.

Item-to-Item Recommendations

"Given a business, recommend similar businesses" — Tier 2 data (7,352 users × 2,218 biz)

Hybrid Similarity Weights (Grid Search)

Combines 3 signals: category overlap, TF-IDF review text, and user co-occurrence (collaborative).



4.11%

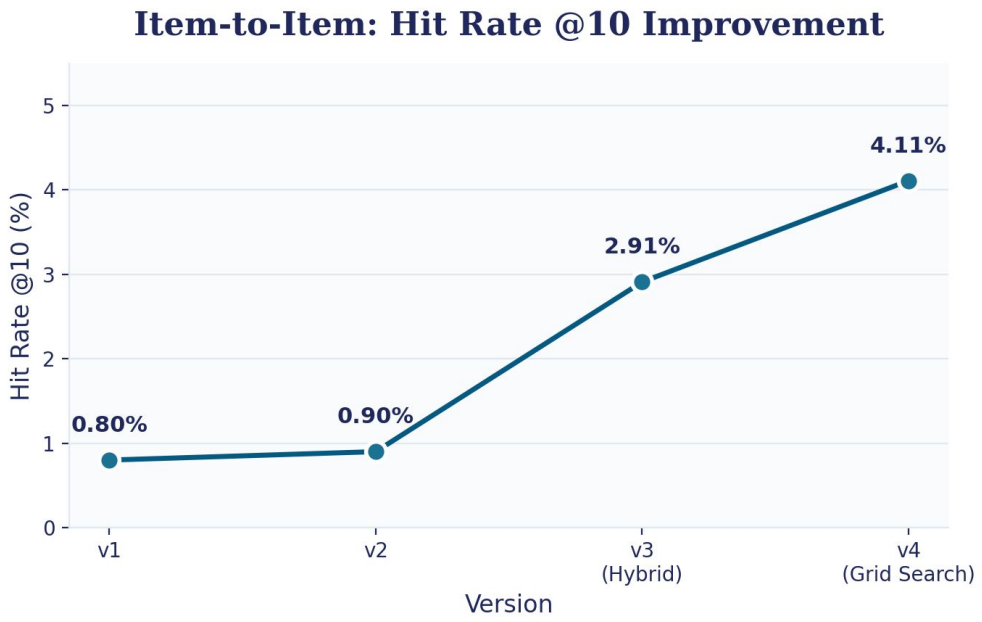
Hit Rate @10

Improved from 2.91% (default weights) to 4.11% after grid search

Collaborative signals dominate category-based content similarity

Item-to-Item: Iteration Journey

"Given a business, recommend similar businesses" — Tier 2 data (7,352 users × 2,218 biz)



Final Model: 4.11% HR@10

Hybrid Similarity Weights (Grid Search)

Collab Co-occurrence: 50%

TF-IDF Text Sim.: 40%

Content (Category): 10%

How Yelp uses this: Displays "Similar Restaurants" on business pages, driving cross-recommendations and increasing discoverability for long-tail venues.

User-to-Business Collaborative Filtering

"Given a user, predict ratings for unvisited businesses" — Tier 3 data (2,279 users × 1,390 biz, 96.8% sparse)

SVD (Matrix Factorization)

Surprise library | 150 factors

lr 0.005, reg 0.2 | 96 param combos

RMSE: 1.10 | MAE: 0.85

Best rating prediction accuracy

Coverage: 16.5% | Diversity: 84.5%

NCF (Neural Collab. Filtering)

GMF + MLP | 241K parameters

~100K training interactions

Coverage: 58.6% | Diversity: 85.3%

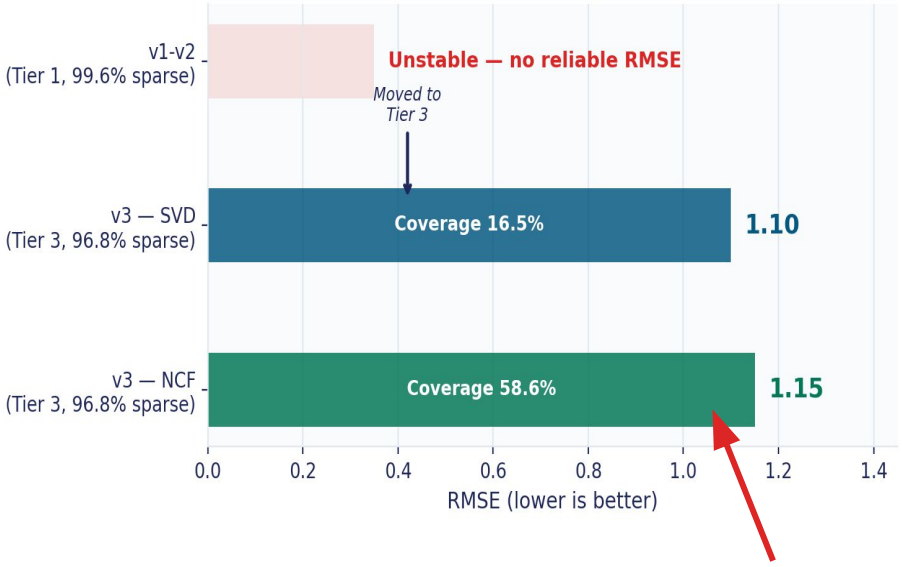
Best catalog coverage & diversity

RMSE: 1.15 (needs 10-50x more data to beat SVD)

User-to-Business: Iteration Journey

"Given a user, predict ratings for unvisited businesses" — Tier 3 data (2,279 users × 1,390 biz)

Iterative Improvement: RMSE progression to stable predictions



Key Insight

Data density was the bottleneck, not model complexity.

v1-v2 used all 22,657 users on T1 (99.64% sparse) — SVD couldn't find stable latent factors. Moving to T3's 2,279 high-activity users cut sparsity to 96.82%, immediately stabilizing predictions.

SVD vs NCF Trade-off

SVD wins on accuracy (RMSE 1.10 vs 1.15) but NCF wins on coverage (58.6% vs 16.5%). With 10-50x more data, NCF would likely surpass SVD — deep models need scale.

RecSys — Two Tasks, Complementary Results

Item-to-Item (Tier 2)

"Similar businesses" recommendations

HR@10: 4.11% (vs. 1.67% popularity baseline)

Best for business pages, "you might also like"

User-to-Business (Tier 3)

Personalized rating predictions

SVD: best RMSE (1.10) | NCF: best coverage (58.6%)

Best for personalized feeds, "recommended for you"

User-to-Business Metrics (Popularity baseline for reference)

Metric	Popularity	SVD	NCF
RMSE	n/a	**1.10**	1.15
Hit Rate @10	**1.67%**	0.75%	1.36%
Coverage	2.1%	16.5%	**58.6%**
Diversity	80.1%	84.5%	**85.3%**

Churn Prediction

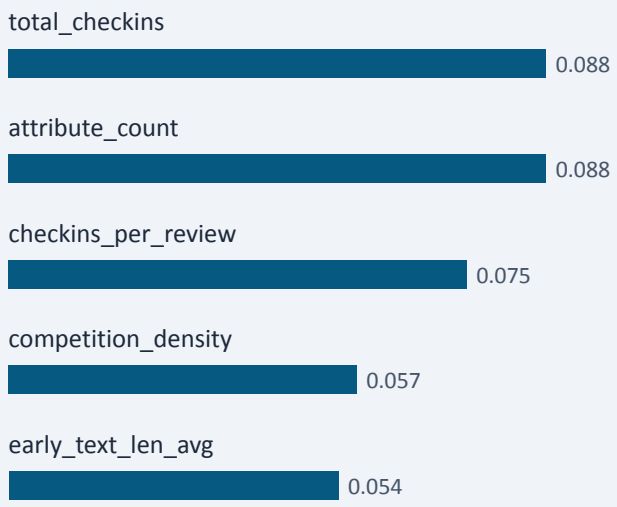
3,769 businesses | 25.4% closed | 30 features | Class-balanced

Model	AUC (CV)	AUC (Test)	F1	Accuracy
Logistic Reg.	0.65	0.68	0.70	0.61
Random Forest	**0.71**	**0.74**	**0.87**	**0.79**
Gradient Boost	0.73	0.73	0.79	0.71

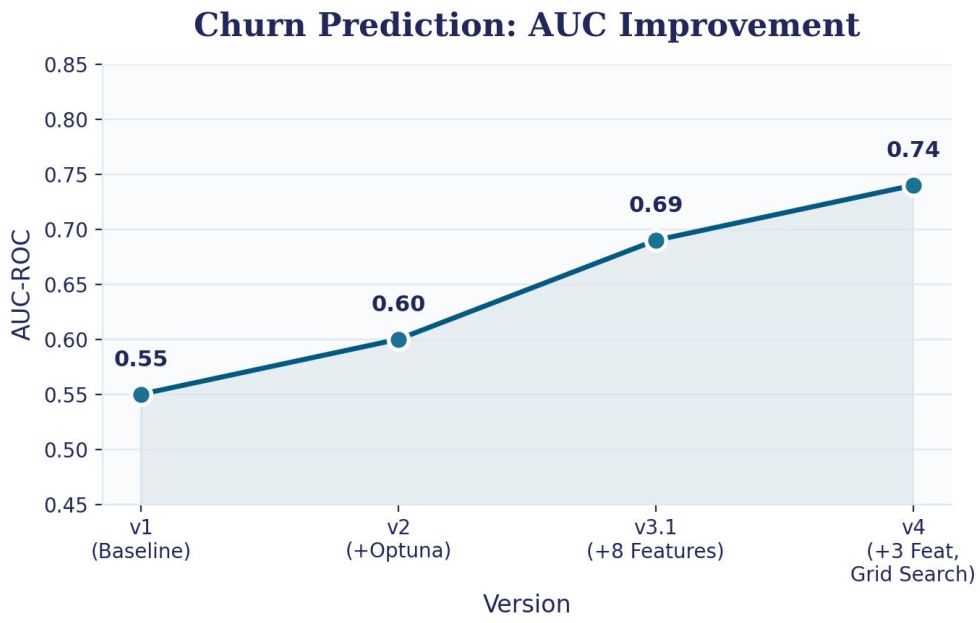
Optimal threshold: 0.53 | Closed-class F1: 0.515

Optuna 100-trial hyperparameter search with 5-fold CV

Top 5 Features



Churn: Iteration Journey & Optimization



Final Model: AUC = 0.74

Random Forest (Optuna tuned)

Key drivers: Foot traffic (checkins), profile completeness, and competitive density.

What it tells us: Survival is tied to engagement and market saturation, not just ratings. How Yelp uses it: Early warning system to flag at-risk businesses for outreach or support programs.

Interpretability & Error Analysis

SHAP Analysis (Random Forest)

Pushes toward "Open":

Higher checkin volumes
More business attributes (profile completeness)
Higher review velocity

Pushes toward "Closed":

High competition density (>85 in 2km)
Low review velocity
High % of 1-star reviews

Error Analysis

False Positives
(predicted open, actually closed)

Decent ratings & engagement but closed anyway. Likely driven by factors outside Yelp data: lease costs, ownership changes, financing.

False Negatives
(predicted closed, actually open)

Show warning signs but survived. Likely due to loyal customer bases, recent pivots, or improvements not captured in early-life data.

Business Insights

1. Collaborative > Content for restaurants

Grid search: collab co-occurrence (0.5) + review text (0.4) >> category matching (0.1). Users who visit similar restaurants share more signal through review language than through categories.

2. Foot traffic is the #1 survival predictor

Total checkins and checkins-per-review are the top churn features. Consistent repeat visitors matter more than star ratings.

3. Profile completeness signals engagement

attribute_count is #2 predictor. Owners who invest in their Yelp profile are more likely to keep their business open.

4. Competition matters but isn't everything

>85 competitors in 2km increases closure risk, but many high-competition businesses thrive through differentiation.

Assumptions & Limitations

Assumptions

- is_open field reliably indicates closure (may miss relocations or name changes)
- First 12-month patterns are predictive of long-term survival
- Tampa Bay scope defined by top 15 FL cities by business count
- User preferences are relatively stable over time (time-based split)

Limitations

- 96.8% sparsity limits collab filtering (explains low hit rates)
- No financial/operational data — hard ceiling on churn AUC (~0.75)
- 100K interactions is low for NCF (needs 10-50x for SVD advantage)
- Results specific to Tampa Bay — needs retraining for other cities
- is_open is a snapshot, not a tracked event — misses reopenings

Conclusion

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RecSys Tasks

Item-to-Item + User-to-Business with tiered data strategy

0.74

Best Churn AUC

30 features, SHAP interpretability + survival analysis

4.11%

Best HR@10

Item-to-Item hybrid with optimized similarity weights

Thank You — Questions?